RAG Learning

4.1 Atomic Structure and the Periodic Table

AQA GCSE Chemistry (8462) — Paper 1

4.1.1 Atoms, Elements and Compounds

> An **atom** is the smallest part of an element that can exist. All substances are made of atoms.
>
> An **element** is a substance made of only **one type of atom**. There are about 100 different elements, and each is shown in the periodic table.
>
> A **compound** contains **two or more elements chemically combined** in fixed proportions.

Every element has a **chemical symbol** of one or two letters — for example O for oxygen, Na for sodium (from the Latin *natrium*) and Fe for iron (*ferrum*). The first letter is always a capital and the second is always lower case: Co is cobalt, but CO is carbon monoxide (a compound of carbon and oxygen). You need to know the names and symbols of the first 20 elements, Groups 1 and 7, and other common elements named in the specification.

Compounds and Chemical Reactions

Compounds are formed from elements by **chemical reactions**. A chemical reaction always involves the formation of one or more **new substances**, and often involves a detectable energy change. Because the atoms in a compound are held together by chemical bonds, a compound can only be separated into its elements by further chemical reactions — *not* by physical methods such as filtration or distillation.

Compounds have properties that are completely different from the elements that made them. Sodium is a soft, highly reactive metal that fizzes in water; chlorine is a toxic green gas. Combined chemically they form sodium chloride — table salt — a white crystalline solid you can safely eat.

© RAG Learning · Not for redistribution

Naming Compounds

Rule	Meaning	Examples
-ide	Compound of a metal and one non-metal (only two elements).	Sodium chlor**ide** NaCl, magnesium ox**ide** MgO, iron sulf**ide** FeS
-ate	Compound containing oxygen as well as two other elements.	Copper sulf**ate** $CuSO_4$, calcium carbon**ate** $CaCO_3$, potassium nitr**ate** KNO_3
Prefixes	Show the number of atoms of an element in a molecule (mono = 1, di = 2, tri = 3, tetra = 4).	Carbon **di**oxide CO_2, carbon **mon**oxide CO, sulfur **tri**oxide SO_3

Word Equations and Symbol Equations

Chemical reactions can be represented by **word equations** or by **symbol equations**. Reactants go on the left of the arrow; products on the right. You should be able to write word equations for reactions in the specification and write formulae and balanced chemical equations for them. Higher-tier students should also be able to write balanced **half equations** and **ionic equations** where appropriate.

```
Word equation:    magnesium + oxygen → magnesium oxide
Symbol equation:  2Mg(s) + O₂(g) → 2MgO(s)

Word equation:    methane + oxygen → carbon dioxide + water
Symbol equation:  CH₄(g) + 2O₂(g) → CO₂(g) + 2H₂O(l)
```

A balanced equation has the **same number of atoms of each element on both sides**. You balance by changing the **large numbers in front** of formulae (coefficients) — never by changing the small subscript numbers inside a formula, as that would change the substance itself.

Worked Example 1 — Balancing an Equation

```
Question: Balance the equation for sodium reacting with water.
          Na + H₂O → NaOH + H₂
```

© RAG Learning · Not for redistribution

```
Step 1:  Count atoms.
         Left:  Na 1, H 2, O 1      Right: Na 1, H 3, O 1
Step 2:  H is unbalanced. Put 2 in front of H₂O and NaOH:
         Na + 2H₂O → 2NaOH + H₂
         Left:  Na 1, H 4, O 2      Right: Na 2, H 4, O 2
Step 3:  Na now unbalanced. Put 2 in front of Na:
         2Na + 2H₂O → 2NaOH + H₂
         Left:  Na 2, H 4, O 2      Right: Na 2, H 4, O 2  ✓

Answer:  2Na(s) + 2H₂O(l) → 2NaOH(aq) + H₂(g)
```

Half Equations and Ionic Equations (Higher)

A **half equation** shows what happens to one species when it gains or loses electrons (e^-). An **ionic equation** shows only the ions that actually take part, leaving out **spectator ions** that are unchanged.

```
Half equations:
  Na → Na⁺ + e⁻              (sodium atom loses one electron)
  Cl₂ + 2e⁻ → 2Cl⁻           (chlorine molecule gains two electrons)

Ionic equation — chlorine displacing bromide ions:
  Full:   Cl₂(aq) + 2KBr(aq) → 2KCl(aq) + Br₂(aq)
  Ionic:  Cl₂(aq) + 2Br⁻(aq) → 2Cl⁻(aq) + Br₂(aq)
          (K⁺ ions are spectator ions — they do not change)
```

> **Exam Tip:** When balancing, check charges as well as atoms in half equations and ionic equations: the **total charge** must be the same on both sides. In `Cl₂ + 2e⁻ → 2Cl⁻`, the left has a total charge of 2– and the right has 2 × (1–) = 2–. Never change the formula of a substance to make an equation balance — H_2O_2 is hydrogen peroxide, not water.

4.1.1.2 Mixtures and Separation Techniques

A **mixture** consists of two or more elements or compounds **not chemically combined** together. The chemical properties of each substance in the mixture are unchanged, and the

> substances can be separated by **physical processes** — no chemical reactions are needed.

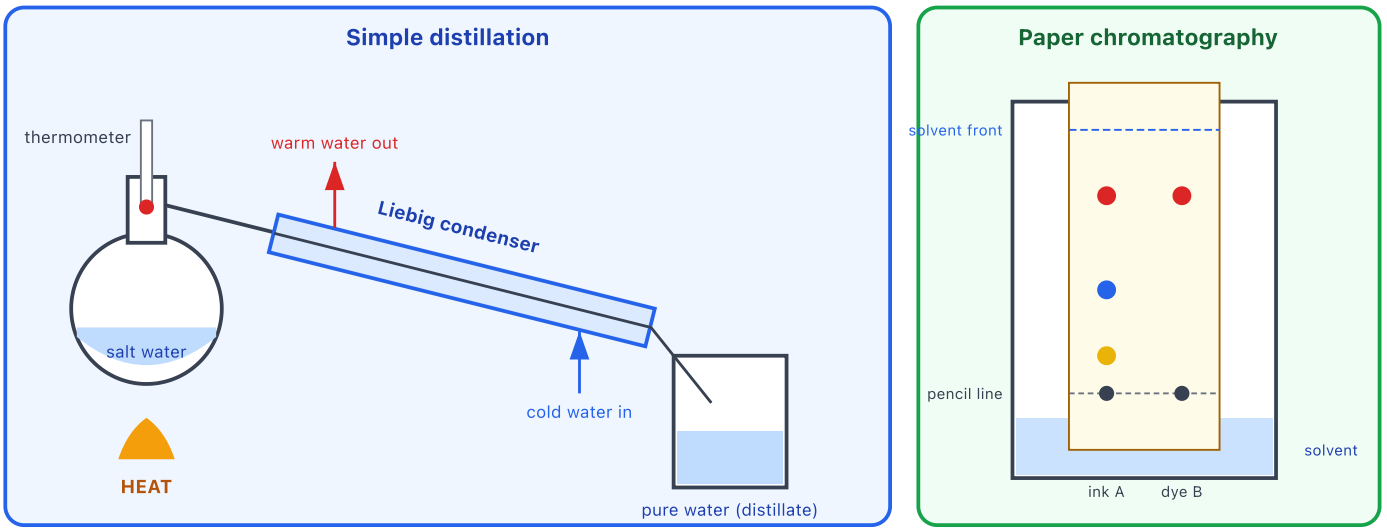
Air (mainly nitrogen, oxygen and argon), sea water (water + dissolved salts) and crude oil (many hydrocarbons) are all mixtures. Because the components are not bonded together, a mixture has no fixed composition and does not have a single sharp melting or boiling point.

Technique	Separates	How it works	Example
Filtration	An insoluble solid from a liquid	Mixture poured through filter paper in a funnel. Liquid (filtrate) passes through the tiny pores; the solid (residue) is trapped.	Sand from water; excess copper oxide from copper sulfate solution.
Crystallisation	A soluble solid from its solution	Solution heated in an evaporating basin to evaporate some solvent until crystals start to form (saturated). Left to cool; crystals form; filtered and dried.	Copper sulfate crystals from copper sulfate solution.
Evaporation	A soluble solid from its solution	Heat to drive off all the solvent. Quick, but the solid may decompose and the crystals are small.	Salt from sea water.
Simple distillation	A solvent from a solution	Solution heated; solvent boils, vapour passes into a condenser, is cooled, and condenses back into a liquid (distillate) which is collected.	Pure water from salt water.
Fractional distillation	Miscible liquids with different boiling points	A fractionating column is placed above the flask. It is hottest at the bottom and coolest at the top. The liquid with the lowest boiling point reaches the top first and distils over first.	Ethanol from water; separating crude oil into fractions.

© RAG Learning · Not for redistribution

Paper chromatography	Soluble substances in a mixture (e.g. dyes)	Substances separate as the solvent (mobile phase) moves up the paper (stationary phase), because each substance has a different attraction to the paper and solubility in the solvent.	Food colourings, inks.

Simple distillation and paper chromatography



In the chromatogram above, ink A separates into three spots, so it is a **mixture** of at least three dyes. Dye B gives a single spot at the same height as the red spot in ink A, so ink A probably contains dye B. Chromatography is covered in detail (including R_f values and pure substances) in topic 4.8 Chemical Analysis.

> **Exam Tip:** When asked to describe a separation, name the technique *and* the property it relies on. Filtration relies on particle size / solubility; distillation relies on **different boiling points**; chromatography relies on different **solubility** in the solvent and attraction to the paper. Draw the chromatography start line in **pencil** (ink would dissolve and run) and keep it **above** the solvent level (otherwise the spots dissolve into the solvent in the beaker).

4.1.1.3 The Development of the Model of the Atom

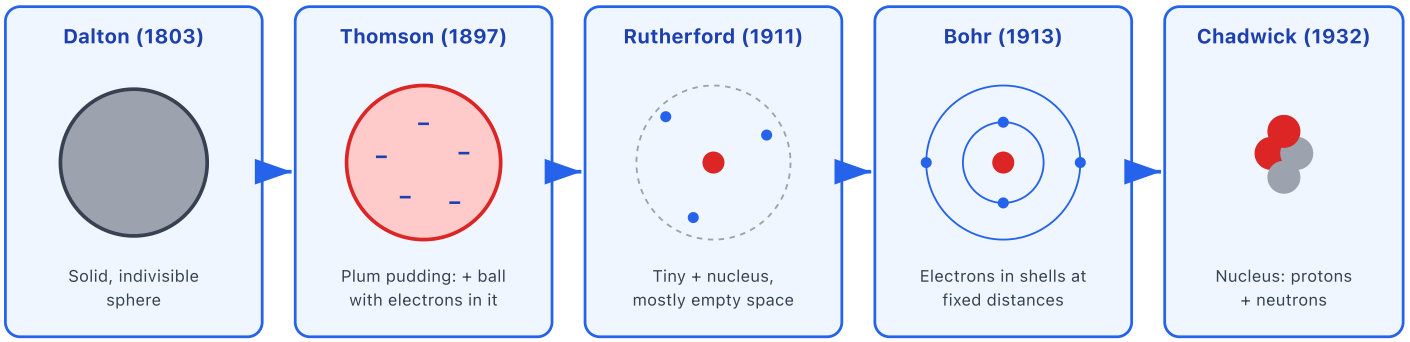
Ideas about atoms have changed many times as new experimental evidence was discovered. This is a classic example of how scientific models are tested and replaced. You need to know the sequence and what evidence caused each change.

Date	Scientist	Model / discovery	Evidence / reason
Early 1800s	**John Dalton**	Atoms are tiny **solid spheres** that cannot be divided. Each element has its own kind of atom.	Elements combine in fixed ratios by mass.
1897	**J.J. Thomson**	Discovered the **electron**. Proposed the **plum pudding model**: a ball of positive charge with negative electrons embedded in it.	Cathode-ray experiments showed atoms contain tiny negative particles, so atoms *can* be divided.
1909–1911	**Ernest Rutherford** (with Geiger and Marsden)	**Nuclear model**: mass concentrated in a tiny, positively charged **nucleus**, surrounded mostly by empty space containing electrons.	The **alpha particle scattering experiment**.
1913	**Niels Bohr**	Electrons orbit the nucleus at specific distances in **shells** (energy levels).	Theoretical calculations agreed with experimental observations (e.g. emission spectra).
c. 1920	Rutherford and others	The positive charge of the nucleus is divided into smaller particles, each with the same amount of positive charge: **protons**.	Later experiments.
1932	**James Chadwick**	Provided evidence for **neutrons** in the nucleus.	About 20 years after the nucleus became an accepted idea; explained why atomic

© RAG Learning · Not for redistribution

			masses were larger than proton counts alone.

How the model of the atom developed



Each model was replaced when new experimental evidence could not be explained by the old one.

The Alpha Particle Scattering Experiment

Rutherford's team fired positively charged **alpha particles** at a very thin sheet of gold foil and recorded where they went. If the plum pudding model were right, the positive charge would be spread thinly through the atom, and all the alpha particles should pass through with only tiny deflections.

Observation	Conclusion
Most alpha particles passed **straight through**.	Most of the atom is **empty space**.
Some were **deflected** through small angles.	The centre of the atom has a **positive charge** that repels the positive alpha particles.
A very few (about 1 in 8000) **bounced back**.	The positive charge and almost all the mass are concentrated in a very **small nucleus**.

Exam Tip: A favourite question: "Compare the plum pudding model with the nuclear model." Give differences in pairs: in the plum pudding model, positive charge is **spread throughout** the atom, whereas in the nuclear model it is **concentrated in the nucleus**; the plum pudding model has **no empty space**, whereas the nuclear model is **mostly empty space**; the plum

© RAG Learning · Not for redistribution

pudding model has electrons **embedded in** the positive charge, whereas the nuclear model has electrons **orbiting** the nucleus (in shells, after Bohr).

4.1.1.4–4.1.1.5 Subatomic Particles, Size and Mass of Atoms

Atoms contain three subatomic particles. The **nucleus** contains **protons** and **neutrons**; the **electrons** are arranged in shells (energy levels) around the nucleus.

Particle	Relative charge	Relative mass	Location
Proton	+1	1	Nucleus
Neutron	0	1	Nucleus
Electron	–1	Very small (about 1/1840)	Shells around nucleus

In an atom, the number of electrons is **equal to** the number of protons in the nucleus, so atoms have **no overall electrical charge**. The number of protons in an atom of an element is its **atomic number**; all atoms of a particular element have the same number of protons, and atoms of different elements have different numbers of protons.

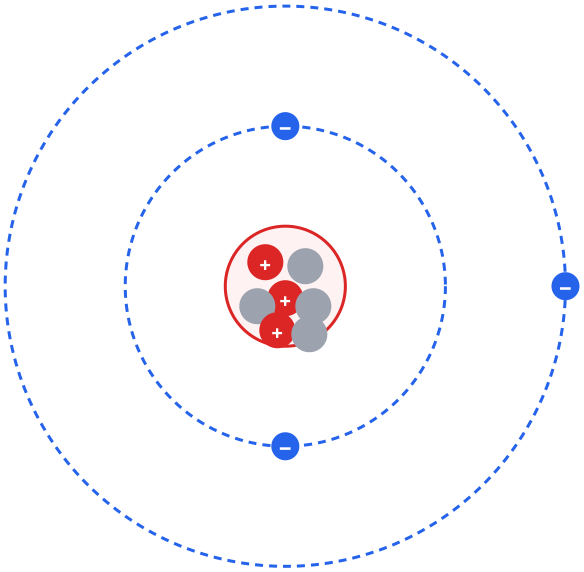
Size of Atoms

- Atoms are very small, having a radius of about **0.1 nm (1×10^{-10} m)**.
- The radius of a nucleus is less than **1/10 000** of that of the atom (about 1×10^{-14} m).
- Almost all of the mass of an atom is in the nucleus.

To picture this: if an atom were the size of a football stadium, the nucleus would be about the size of a pea on the centre spot. You should be able to relate size and scale of atoms to objects in the physical world, and use standard form and prefixes (1 nm = 1×10^{-9} m).

© RAG Learning · Not for redistribution

Nuclear model of a lithium-7 atom (not to scale)



Key

- proton (3) — charge +1
- neutron (4) — charge 0
- electron (3) — charge −1

Mass number = 3 + 4 = 7
Electronic structure: 2,1

Atomic radius ≈ 1×10^{-10} m (0.1 nm)
Nuclear radius ≈ 1×10^{-14} m
Nucleus holds almost all the mass

Mass Number, Atomic Number and Isotopes

Atomic number = number of protons (also the number of electrons in a neutral atom).
Mass number = total number of protons **and** neutrons in the nucleus.
Number of neutrons = mass number − atomic number.

Atoms are often shown in nuclide notation, with the mass number written at the top left and the atomic number at the bottom left of the symbol: for sodium, $^{23}_{11}$**Na** has 11 protons, 11 electrons and 23 − 11 = **12 neutrons**.

Isotopes are atoms of the same element with the **same number of protons** but a **different number of neutrons** (so different mass numbers). Isotopes of an element have the same chemical properties because they have the same electronic structure.

Isotope	Protons	Neutrons	Electrons	Notes
$^{12}_{6}$C (carbon-12)	6	6	6	Most common carbon isotope (~99%).

© RAG Learning · Not for redistribution

$^{14}_{6}C$ (carbon-14)	6	8	6	Radioactive — used in carbon dating.
$^{35}_{17}Cl$ (chlorine-35)	17	18	17	About 75% of natural chlorine.
$^{37}_{17}Cl$ (chlorine-37)	17	20	17	About 25% of natural chlorine.
$^{1}_{1}H$ (hydrogen)	1	0	1	The only atom with no neutrons.
$^{2}_{1}H$ (deuterium)	1	1	1	"Heavy hydrogen".

Ions

Atoms can gain or lose electrons to form charged particles called **ions**. The number of protons never changes. For example, $^{27}_{13}Al^{3+}$ has 13 protons, 14 neutrons and 13 − 3 = **10 electrons**; $^{32}_{16}S^{2-}$ has 16 protons, 16 neutrons and 16 + 2 = **18 electrons**.

> **Exam Tip:** When calculating subatomic particles in an ion, work out protons first (= atomic number), then neutrons (mass number – atomic number), then electrons last: **subtract** the charge for a positive ion, **add** it for a negative ion. A very common mistake is to change the number of protons when an ion forms — protons never change in a chemical reaction.

4.1.1.6 Relative Atomic Mass

> The **relative atomic mass (A_r)** of an element is an **average** value that takes account of the **abundance of the isotopes** of the element. It is measured relative to one-twelfth of the mass of a carbon-12 atom, so it has no units.

Most elements exist naturally as a mixture of isotopes. This is why many A_r values on the periodic table are not whole numbers (Cl = 35.5, Cu = 63.5).

© RAG Learning · Not for redistribution

> A_r = Σ(isotope mass × abundance) ÷ total abundance
>
> If abundances are given as percentages, the total abundance is 100.

Worked Example 2 — Relative Atomic Mass of Chlorine

```
Question: Chlorine is 75% chlorine-35 and 25% chlorine-37.
          Calculate the relative atomic mass of chlorine.

Formula: Ar = (mass₁ × abundance₁ + mass₂ × abundance₂) ÷ 100
Working: Ar = (35 × 75 + 37 × 25) ÷ 100
           = (2625 + 925) ÷ 100
           = 3550 ÷ 100
           = 35.5

Answer:  Ar(Cl) = 35.5
```

Worked Example 3 — Relative Atomic Mass of Copper

```
Question: A sample of copper contains 69.2% copper-63 and 30.8%
          copper-65. Calculate the relative atomic mass to 1 d.p.

Working: Ar = (63 × 69.2 + 65 × 30.8) ÷ 100
           = (4359.6 + 2002.0) ÷ 100
           = 6361.6 ÷ 100
           = 63.616

Answer:  Ar(Cu) = 63.6 (to 1 d.p.)

Sense check: the answer must lie between 63 and 65, and closer
to 63 because copper-63 is the more abundant isotope. ✓
```

Worked Example 4 — Finding an Unknown Abundance

```
Question: Boron has two isotopes, boron-10 and boron-11, and
          Ar = 10.8. Calculate the percentage of boron-10.

Let % of boron-10 = x, so % of boron-11 = 100 − x
```

© RAG Learning · Not for redistribution

```
          10.8 = (10x + 11(100 − x)) ÷ 100
          1080 = 10x + 1100 − 11x
          1080 = 1100 − x
             x = 20

Answer:  20% boron-10, 80% boron-11
```

Exam Tip: Always sanity-check an A_r answer: it must lie between the lightest and heaviest isotope masses, and nearer the most abundant one. If the abundances are given as numbers that do not add up to 100 (e.g. from a mass spectrum with relative heights 3 and 1), divide by their **total**, not by 100.

4.1.1.7 Electronic Structure

Electrons occupy the **lowest available energy levels** (innermost available shells) first. For the first 20 elements:

- the **first shell** holds up to **2** electrons
- the **second shell** holds up to **8** electrons
- the **third shell** holds **8** electrons before the fourth shell starts to fill (for elements 1–20)

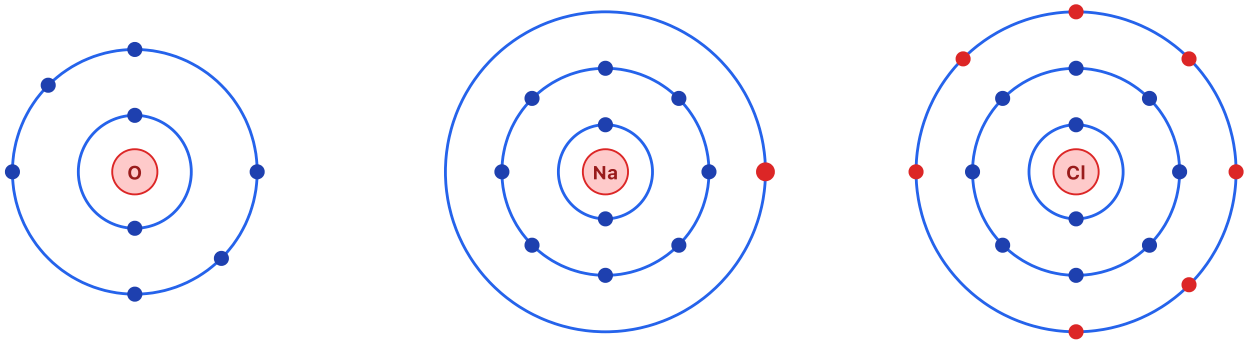
The electronic structure of an atom can be written as numbers separated by commas (e.g. sodium is **2,8,1**) or drawn as a diagram of concentric circles with electrons shown as dots or crosses.

Element	Atomic number	Electronic structure	Element	Atomic number	Electronic structure
H	1	1	Na	11	2,8,1
He	2	2	Mg	12	2,8,2
Li	3	2,1	Al	13	2,8,3
Be	4	2,2	Si	14	2,8,4

© RAG Learning · Not for redistribution

B	5	2,3	P	15	2,8,5
C	6	2,4	S	16	2,8,6
N	7	2,5	Cl	17	2,8,7
O	8	2,6	Ar	18	2,8,8
F	9	2,7	K	19	2,8,8,1
Ne	10	2,8	Ca	20	2,8,8,2

Electronic structures drawn as shell diagrams



Oxygen: 2,6
Group 6, Period 2

Sodium: 2,8,1
1 outer electron → Group 1

Chlorine: 2,8,7
7 outer electrons → Group 7

Link to the periodic table: the **number of shells** containing electrons equals the **period** (row) number. The **number of electrons in the outer shell** equals the **group** number (for Groups 1–7; Group 0 elements have a full outer shell).

Exam Tip: For the first 20 elements, the fourth shell starts filling at potassium (2,8,8,1) even though the third shell could, in theory, hold more. Do not write potassium as 2,8,9. Draw electrons in **pairs** around the outer shell when possible — it makes bonding diagrams much easier later.

4.1.2 The Periodic Table

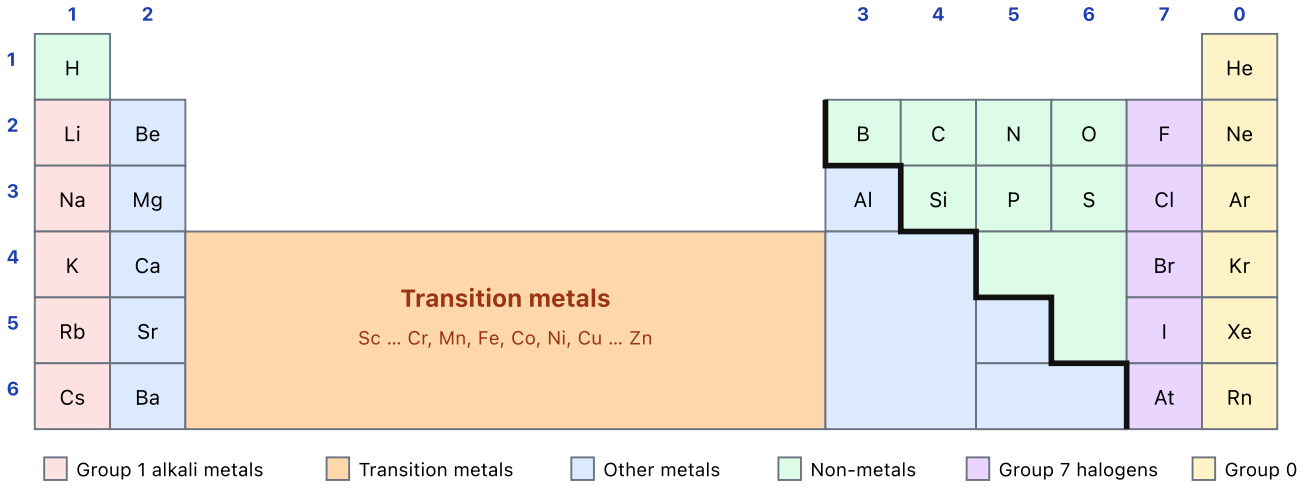
© RAG Learning · Not for redistribution

4.1.2.1 The Periodic Table

> The elements in the periodic table are arranged in order of **atomic (proton) number** so that elements with **similar properties** are in columns, known as **groups**. The horizontal rows are called **periods**. The table is called a *periodic* table because similar properties occur at regular intervals.

Elements in the same group have the **same number of electrons in their outer shell**, and this gives them similar chemical properties — chemical reactions only involve the outer electrons. You should be able to explain how the position of an element in the periodic table is related to the arrangement of electrons in its atoms, and hence to its atomic number, and to predict possible reactions and probable reactivity of elements from their positions.

The periodic table — groups, periods, metals and non-metals



The thick stepped line separates metals (left) from non-metals (right). Hydrogen is a non-metal.

4.1.2.2 Development of the Periodic Table

Before the discovery of protons, neutrons and electrons, scientists attempted to classify the elements by arranging them in order of their **atomic weights**.

- **Early periodic tables** were incomplete. Some elements were placed in inappropriate groups if the strict order of atomic weights was followed. John Newlands' "law of octaves" (1864) noticed that every eighth element had similar properties, but his table broke down after calcium and grouped some very different elements together (e.g. metals with non-metals), so it was not widely accepted.

© RAG Learning · Not for redistribution

- **Dmitri Mendeleev** (1869) overcame some of these problems by:
  - **leaving gaps** for elements that he thought had not yet been discovered;
  - in some places **changing the order** based on atomic weights so that elements lined up with others of similar properties (e.g. he placed iodine after tellurium even though tellurium has the higher atomic weight).
- Elements with properties predicted by Mendeleev were later discovered and **filled the gaps** — for example gallium (1875) and germanium (1886), whose properties closely matched his predictions for "eka-aluminium" and "eka-silicon". This strongly supported his table.
- Knowledge of **isotopes** made it possible to explain why the order based on atomic weights was not always correct: atomic weight depends on the number of neutrons, but an element's chemistry depends on its proton number. Ordering by **atomic number** places every element correctly.

> **Exam Tip:** A common 4–6 mark question asks you to compare Mendeleev's table with the modern periodic table. Key points: Mendeleev ordered by **atomic weight** (modern: atomic number); he left **gaps** (modern table has no gaps for elements up to 118); he had **no Group 0** (noble gases were undiscovered because they are so unreactive); and he swapped some pairs (Te/I) — later justified by the discovery of isotopes and atomic number.

4.1.2.3 Metals and Non-metals

> **Metals** are elements that react to form **positive ions**. They are found to the left and towards the bottom of the periodic table. The majority of elements are metals.
>
> **Non-metals** are elements that **do not form positive ions**. They are found towards the right and top of the periodic table.

Whether an element forms positive ions depends on its electronic structure. Metal atoms have only a few outer electrons (1, 2 or 3), so it takes relatively little energy for them to **lose** these electrons and achieve the stable electronic structure of a noble gas. Non-metal atoms have nearly full outer shells, so they tend to **gain** electrons (forming negative ions) or **share** electrons (covalent bonds).

© RAG Learning · Not for redistribution

Property	Metals (typical)	Non-metals (typical)
Appearance	Shiny (lustrous) when freshly cut	Dull
Melting / boiling points	Mostly high	Mostly low (many are gases at room temperature)
Electrical and thermal conductivity	Good conductors	Poor conductors / insulators (graphite is an exception)
Malleability	Malleable and ductile	Brittle when solid
Density	Usually high	Usually low
Oxides	Basic oxides (react with acids)	Acidic oxides (form acidic solutions)
Ions formed	Positive ions (cations)	Negative ions (anions), or share electrons

4.1.2.4 Group 0 — The Noble Gases

> The elements in Group 0 are called the **noble gases**: helium, neon, argon, krypton, xenon and radon. They are **unreactive** and do not easily form molecules because their atoms have **stable arrangements of electrons**. The noble gases have **eight electrons in their outer shell**, except for helium, which has only two.

Because noble gases exist as single atoms (they are **monatomic**), the only forces between them are very weak intermolecular forces, so their boiling points are very low. The boiling points of the noble gases **increase with increasing relative atomic mass** (going down the group), because the atoms get larger and the forces between them become stronger.

© RAG Learning · Not for redistribution

Noble gas	Symbol	Electronic structure	Boiling point (°C)	Uses
Helium	He	2	−269	Balloons and airships (low density, non-flammable)
Neon	Ne	2,8	−246	Advertising signs (glows red when electricity passes through)
Argon	Ar	2,8,8	−186	Filament lamps and welding (inert atmosphere stops metal reacting with oxygen)
Krypton	Kr	2,8,18,8	−153	Some lasers, photographic flash lamps
Xenon	Xe	…,8	−108	Car headlamps, flash lamps
Radon	Rn	…,8	−62	Radioactive — a health hazard in some homes

Exam Tip: To predict a property of a noble gas from a trend, look at the pattern in the data: e.g. if asked for the boiling point of xenon given krypton (−153 °C) and radon (−62 °C), a value roughly halfway (≈ −108 °C) is ideal. Explain unreactivity in terms of **full outer shells** — noble gases have no tendency to lose, gain or share electrons.

4.1.2.5 Group 1 — The Alkali Metals

The elements in Group 1 are known as the **alkali metals** (lithium, sodium, potassium, rubidium, caesium, francium). They have characteristic properties because of the **single electron in their outer shell**. They react with water to form alkaline hydroxide solutions.

Alkali metals are soft (they can be cut with a knife), have low densities (Li, Na and K float on water) and have relatively low melting points for metals. They are stored under oil because they react

© RAG Learning · Not for redistribution

rapidly with oxygen and water vapour in the air. When freshly cut they are shiny, but they quickly tarnish as a layer of oxide forms.

Reactions of Group 1 Metals

Reaction with	General equation	Example (sodium)
Water	metal + water → metal hydroxide + hydrogen	$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$
Oxygen	metal + oxygen → metal oxide	$4Na(s) + O_2(g) \rightarrow 2Na_2O(s)$
Chlorine	metal + chlorine → metal chloride	$2Na(s) + Cl_2(g) \rightarrow 2NaCl(s)$

Metal	Observations with water	Melting point (°C)
Lithium	Floats, fizzes steadily, moves slowly around the surface, gradually disappears.	181
Sodium	Floats, melts into a silvery ball (reaction is exothermic and Na has a low melting point), fizzes rapidly, whizzes around the surface.	98
Potassium	Floats, melts, fizzes violently, the hydrogen produced ignites with a **lilac flame**, may spit and crackle.	63
Rubidium / caesium	Explode on contact with water — not used in schools.	39 / 28

Universal indicator added to the water turns **purple** because an alkaline metal hydroxide solution is produced (pH 13–14). The compounds formed are all **ionic** and are white solids that dissolve in water to form colourless solutions.

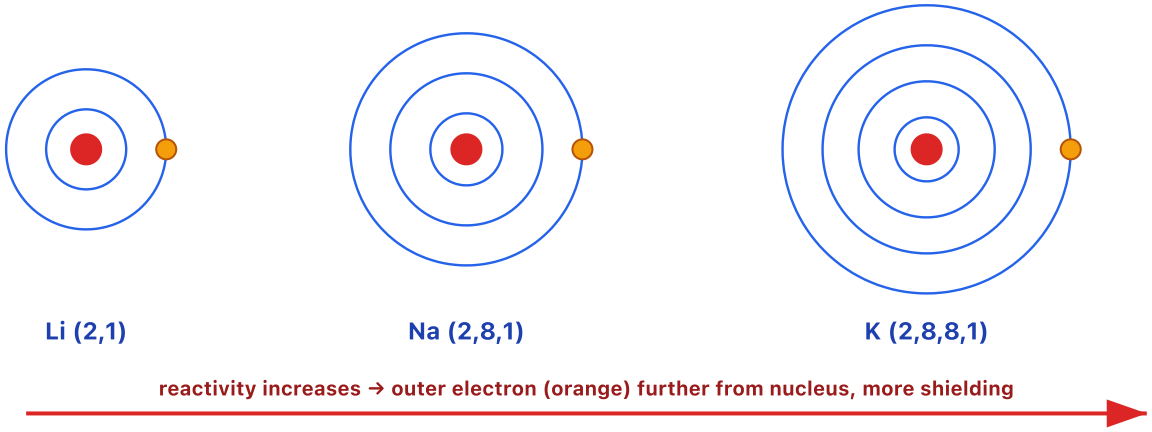
Explaining the Trend in Reactivity

© RAG Learning · Not for redistribution

Reactivity increases going down Group 1. In all their reactions, Group 1 atoms **lose their single outer electron** to form a **1+ ion** (e.g. Na → $Na^+ + e^-$). Going down the group:

- the atoms get **larger** — more shells of electrons;
- the outer electron is **further from the nucleus** and more **shielded** by inner shells;
- so there is **weaker electrostatic attraction** between the nucleus and the outer electron;
- so the outer electron is **lost more easily**.

Going down Group 1: bigger atoms, outer electron lost more easily



Exam Tip: For a full-mark explanation of the Group 1 trend you need **four** linked points: more shells → outer electron further from nucleus (plus more shielding) → weaker attraction to the nucleus → outer electron lost more easily. Do not say "the outer electron is *easier to gain*" — Group 1 metals **lose** electrons. Avoid saying "the bonds are weaker" — it is the **attraction between the nucleus and the outer electron** that is weaker.

4.1.2.6 Group 7 — The Halogens

The elements in Group 7 are known as the **halogens** (fluorine, chlorine, bromine, iodine, astatine). They have similar reactions because they all have **seven electrons in their outer shell**. The halogens are non-metals and consist of **diatomic molecules** made of pairs of atoms (F_2, Cl_2, Br_2, I_2).

© RAG Learning · Not for redistribution

Halogen	Formula	State at room temp.	Colour	Melting point (°C)	Boiling point (°C)
Fluorine	F_2	Gas	Pale yellow	−220	−188
Chlorine	Cl_2	Gas	Green (yellow-green)	−101	−34
Bromine	Br_2	Liquid	Red-brown (orange vapour)	−7	59
Iodine	I_2	Solid	Grey-black (purple vapour)	114	184

Going down Group 7, the halogens have **higher relative molecular masses**, so the **melting and boiling points increase**: the larger molecules have stronger intermolecular forces, which need more energy to overcome. The colours also become darker.

Reactions of the Halogens

- **With metals:** halogens react to form **ionic** compounds (metal halides) in which the halide ions carry a **1− charge**. e.g. $2Na + Cl_2 \rightarrow 2NaCl$; $2Fe + 3Cl_2 \rightarrow 2FeCl_3$.
- **With non-metals:** halogens form **covalent** molecular compounds, e.g. $H_2 + Cl_2 \rightarrow 2HCl$ (hydrogen chloride gas, which dissolves in water to form hydrochloric acid).
- **Displacement reactions:** a **more reactive halogen can displace a less reactive halogen** from an aqueous solution of its salt.

Explaining the Trend in Reactivity

> **Reactivity decreases going down Group 7.** Halogen atoms react by **gaining one electron** to form a **1− ion** with a full outer shell. Going down the group:
> • the atoms get **larger** — more shells of electrons;
> • the outer shell is **further from the nucleus** and more **shielded**;
> • so there is **weaker attraction** from the nucleus for an incoming electron;
> • so an electron is **gained less easily**.

© RAG Learning · Not for redistribution

Displacement Reactions

Halogen added ↓ / Salt solution →	Potassium chloride (KCl)	Potassium bromide (KBr)	Potassium iodide (KI)
Chlorine water	No reaction	Orange solution (Br_2 formed)	Brown solution (I_2 formed)
Bromine water	No reaction	No reaction	Brown solution (I_2 formed)
Iodine solution	No reaction	No reaction	No reaction

```
Chlorine + potassium bromide → potassium chloride + bromine
  Cl₂(aq) + 2KBr(aq) → 2KCl(aq) + Br₂(aq)

Ionic equation (Higher):  Cl₂ + 2Br⁻ → 2Cl⁻ + Br₂

Half equations (Higher):  Cl₂ + 2e⁻ → 2Cl⁻      (chlorine is reduced – gains ele
                          2Br⁻ → Br₂ + 2e⁻      (bromide is oxidised – loses ele
```

Exam Tip: Students often reverse the Group 1 and Group 7 trends. Remember: Group 1 reactivity **increases** down the group (electron **lost** more easily); Group 7 reactivity **decreases** down the group (electron **gained** less easily). The underlying reason is the same — larger atoms, weaker attraction between nucleus and outer shell — but the effect is opposite because one group loses and the other gains electrons. In displacement questions, use **halide** (bromide, iodide) for the ions and **halogen** (bromine, iodine) for the element.

4.1.3 Properties of Transition Metals (Triple Only)

4.1.3.1 Comparison with Group 1 Elements

The transition elements are metals found in the central block of the periodic table, between Groups 2 and 3. They have similar properties to each other and some special properties, but they

are quite different from the Group 1 metals. You should be able to describe the difference compared with Group 1 in melting points, densities, strength, hardness and reactivity with oxygen, water and halogens. Examples include **Cr, Mn, Fe, Co, Ni, Cu**.

Property	Group 1 (alkali metals)	Transition metals
Melting point	Low (e.g. Na 98 °C, K 63 °C)	High (e.g. Fe 1538 °C, Cr 1907 °C) — except mercury
Density	Low (Li, Na, K float on water)	High (e.g. Fe 7.9 g/cm³, Cu 8.9 g/cm³)
Strength & hardness	Soft — can be cut with a knife; weak	Hard and strong — used for construction, tools
Reactivity with oxygen	Tarnish rapidly in air; stored under oil	React slowly or not at all at room temperature (iron rusts slowly in the presence of water)
Reactivity with water	Vigorous reaction, forming hydroxide + hydrogen	Little or no reaction with cold water
Reactivity with halogens	Vigorous, forming white 1+ halides	React less vigorously (often need heating), forming coloured halides
Ions formed	Only 1+ ions	Ions with **different charges** (e.g. Fe^{2+}, Fe^{3+})
Colour of compounds	White solids; colourless solutions	**Coloured** compounds

4.1.3.2 Typical Properties of Transition Metals

Many transition elements:

- have **ions with different charges**;
- form **coloured compounds**;
- are useful as **catalysts**.

Metal	Common ions	Colour of compounds / solutions	Catalytic use
Iron (Fe)	Fe^{2+}, Fe^{3+}	Fe^{2+}: pale green; Fe^{3+}: orange-brown (rust colour)	Iron catalyst in the **Haber process** (making ammonia)
Copper (Cu)	Cu^{+}, Cu^{2+}	Cu^{2+}: blue (copper sulfate solution)	Catalyses some oxidation reactions
Nickel (Ni)	Ni^{2+}	Green	Hydrogenation of vegetable oils to make margarine
Manganese (Mn)	Mn^{2+}, Mn^{4+}, Mn^{7+}	MnO_4^- (manganate(VII)): purple	Manganese(IV) oxide catalyses decomposition of hydrogen peroxide
Chromium (Cr)	Cr^{2+}, Cr^{3+}	Cr^{3+}: green; dichromate: orange	Used in chrome plating and stainless steel
Cobalt (Co)	Co^{2+}, Co^{3+}	Co^{2+}: pink (aqueous), blue (anhydrous cobalt chloride)	Cobalt chloride paper tests for water (blue → pink)

Because transition metals can form ions with different charges, their compound names include a Roman numeral: iron(II) chloride is $FeCl_2$ and iron(III) chloride is $FeCl_3$; copper(II) oxide is CuO. Metal hydroxide precipitates of transition metal ions have characteristic colours, which is used in chemical analysis (topic 4.8): Cu^{2+} gives a **blue** precipitate, Fe^{2+} a **green** precipitate and Fe^{3+} a **brown** precipitate with sodium hydroxide solution.

> **Exam Tip:** If asked to "give **two** properties of transition metals that Group 1 metals do not have", choose from: form **coloured compounds**, form **ions with different charges**, act as **catalysts**, have **higher melting points**, have **higher densities**, are **harder/stronger**. Be specific — "they are stronger" scores, "they are better" does not. Always name a correct example of a transition-metal catalyst with its process (iron → Haber process; nickel → hydrogenation).

© RAG Learning · Not for redistribution

Summary Checklist for 4.1 Atomic Structure and the Periodic Table

- Define atom, element, compound and mixture; use chemical symbols and name compounds (-ide, -ate).
- Write word equations and balanced symbol equations (Higher: half equations and ionic equations).
- Describe and explain filtration, crystallisation, evaporation, simple distillation, fractional distillation and chromatography.
- Describe how the atomic model developed: Dalton → Thomson (plum pudding) → Rutherford (alpha scattering, nucleus) → Bohr (shells) → protons → Chadwick (neutrons).
- Recall the relative charges and masses of protons, neutrons and electrons.
- Recall atomic radius ≈ 0.1 nm and nuclear radius < 1/10 000 of the atom.
- Calculate the numbers of protons, neutrons and electrons in atoms and ions from atomic and mass numbers.
- Define isotopes and calculate relative atomic mass from isotopic abundances.
- Write and draw electronic structures for the first 20 elements and link them to group and period.
- Explain how the periodic table is arranged, and how Mendeleev's table differed from earlier and modern tables.
- Distinguish metals (form positive ions) from non-metals and explain this using electronic structure.
- Explain the unreactivity of the noble gases and the trend in their boiling points.
- Describe the reactions of Group 1 metals with water, oxygen and chlorine and explain the trend in reactivity.
- Describe the properties and reactions of the halogens, including displacement reactions, and explain the trend in reactivity.
- (Triple) Compare transition metals with Group 1 metals and describe their typical properties: ions of different charges, coloured compounds, catalysts.

© RAG Learning · Not for redistribution